# Hao-Jun Michael Shi

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## Education

**Northwestern University** — **Evanston, IL**
*Ph.D. in Industrial Engineering and Management Sciences* — *2016–2021*
Advisor: Prof. Jorge Nocedal
*Dissertation: Methods for Stochastic, Noisy, and Derivative-Free Optimization*

**Kellogg School of Management at Northwestern University** — **Evanston, IL**
*Management for Scientists and Engineers Certificate* — *2020*

**Northwestern University** — **Evanston, IL**
*M.S. in Industrial Engineering and Management Sciences* — *2016–2017*

**University of California, Los Angeles** — **Los Angeles, CA**
*B.S. in Applied Mathematics* — *2012–2016*
College and Departmental Honors

## Research Interests

- Numerical Optimization
- Deep Learning
- Mathematical Software
- Scientific Computing

## Awards

2024 1st Place (External Tuning Ruleset), MLCommons AlgoPerf: Training Algorithms Benchmark Competition
2019 International Conference on Machine Learning Top 5% Reviewer
2017, 2018 IEMS Departmental Service Award
2016, 2017 NSF Graduate Research Fellowship: Honorable Mention
2016 Walter P. Murphy Fellowship

## Industry Experience

**Meta Platforms** — **Menlo Park, CA**
*Research Scientist* — *2021–Current*
AI and Systems Co-Design Team, Infrastructure (2021–2025)
Kernels & Optimizations Team, Meta Superintelligence Labs (2025–2026)
FAIR, Meta Superintelligence Labs (2026–Current)

**Facebook** — **Menlo Park, CA**
*Research Intern* — *2019*
Advisor: Dheevatsa Mudigere
Peers: Maxim Naumov and Jiyan Yang

## Conference Proceedings

1. R. Eschenhagen, A. Cai, T.-H. Lee, H.-J.M. Shi. "Clarifying Shampoo: Adapting Spectral Descent to Stochasticity and the Parameter Trajectory". In Progress, 2026. [ArXiv].

2. H. Naganuma, S. Gupta, Y. Briki, I. Mitliagkas, I. Rish, P. Raman, H.-J.M. Shi. "Adaptive Batch Sizes Using Non-Euclidean Gradient Noise Scales for Stochastic Sign and Spectral Descent". International Conference on Machine Learning (ICML), Seoul, South Korea, July 2026. [ArXiv].
3. A. Defazio, K. Mishchenko, P. Raman, H.-J.M. Shi, L. Xiao. "Smoothing DiLoCo with Primal Averaging for Faster Training of LLMs". In Progress, 2025. [ArXiv].
4. R. Eschenhagen, A. Defazio, T.-H. Lee, R.E. Turner, H.-J.M. Shi. "Purifying Shampoo: Investigating Shampoo's Heuristics by Decomposing its Preconditioner". Neural Information Processing Systems (NeurIPS), San Diego, CA, December 2025. [ArXiv].
5. H.-J.M. Shi, D. Mudigere, M. Naumov, and J. Yang. "Compositional Embeddings Using Complementary Partitions for Memory-Efficient Recommendation Systems". KDD, Virtual Conference, August 2020. [KDD].
6. R. Bollapragada, D. Mudigere, J. Nocedal, H.-J.M. Shi, and P.T.P. Tang. "A Progressive Batching L-BFGS Method for Machine Learning". International Conference on Machine Learning (ICML), Stockholm, Sweden, July 2018. [ICML]
7. H.-J.M. Shi, M. Case, X. Gu, S. Tu, and D. Needell. "Methods for Quantized Compressed Sensing". Proc. Information Theory and Applications (ITA), La Jolla, CA, Jan. 2016. [ITA].

## Journal Publications

1. H.-J.M. Shi, Y. Xie, M.Q. Xuan, and J. Nocedal. "Adaptive Finite-Difference Interval Estimation for Noisy Derivative-Free Optimization". SIAM Journal of Scientific Computing, 44(4), A2302-A2321, 2022. [ArXiv].
2. H.-J.M. Shi, M.Q. Xuan, F. Oztoprak, and J. Nocedal. "On the Numerical Performance of Derivative-Free Optimization Methods Based on Finite-Difference Approximations". Optimization Methods and Software, 1-23, 2022. [ArXiv].
3. H.-J.M. Shi, Y. Xie, R. Byrd, and J. Nocedal. "A Noise-Tolerant Quasi-Newton Algorithm for Unconstrained Optimization". SIAM Journal of Optimization, 32(1), 29-55, 2022. [ArXiv].
4. J. Luo, K. Shapiro, H.-J.M. Shi, Q. Yang, and K. Zhu. "Practical Algorithms for Learning Near-Isometric Linear Embeddings". SIAM Undergraduate Research Online, vol. 9, 2016. [SIURO].

## Technical Reports

1. D. Kallusky, V. Rao, V. Nandavanam, H.-J.M. Shi. "SNOO: Step-K Nesterov Outer Optimizer - The Surprising Effectiveness of Nesterov Momentum Applied to Pseudo-Gradients". Preprint, 2025. [ArXiv].
2. H.-J.M. Shi, T.-H. Lee, S. Iwasaki, J. Gallego-Posada, Z. Li, K. Rangadurai, D. Mudigere, and M. Rabbat. "A Distributed Data-Parallel PyTorch Implementation of the Distributed Shampoo Optimizer for Training Neural Networks At-Scale". [ArXiv].
3. M. Naumov, D. Mudigere, H.-J.M. Shi, J. Huang, N. Sundaraman, J. Park, X. Wang, U. Gupta, C.-J. Wu, A.G. Azzolini, D. Dzhulgakov, A. Mallevich, I. Cherniavskii, Y. Lu, R. Krishnamoorthi, A. Yu, V. Kondratenko, S. Pereira, X. Chen, W. Chen, V. Rao, B. Jia, L. Xiong, M. Smelyanskiy. "Deep Learning Recommendation Model for Personalization and Recommendation Systems". Preprint. [ArXiv].
4. H.-J.M. Shi, S. Tu, Y. Xu, and W. Yin. "A Primer on Coordinate Descent Algorithms". Preprint. [ArXiv].
5. X. Gu, S. Tu, H.-J.M. Shi, M. Case, D. Needell, and Y. Plan. "Optimizing Quantization for Lasso Recovery". IEEE Signal Processing Letters, vol. 25, issue 1, Jan. 2018. [IEEE].
6. C. Abrahamson, H.-J.M. Shi, and B. Yang. "Ground Motion Prediction Equations for Arias Intensity Consistent with the NGA-West2 Ground Motion Models". Pacific Earthquake Engineering Research (PEER) Report, July 2016. [PEER].

## Software

- **PyTorch Distributed Shampoo**: A PyTorch implementation of the Distributed Shampoo optimizer compatible with DDP and FSDP. [Code].

- **PyTorch-LBFGS**: A modular PyTorch implementation of stochastic L-BFGS. [Code].

## Presentations

1. "Beyond AdamW: Developments in Training Algorithms for Deep Learning". Optimization Group Seminar, Colorado School of Mines, Remote, Apr. 2026.
2. "Clarifying Shampoo: Adapting Spectral Descent to Stochasticity and the Parameter Trajectory". INFORMS Optimization Society Conference 2026, Atlanta, Georgia, Mar. 2026.
3. "Re-Thinking DiLoCo through Generalized Primal Averaging for Distributed Training". Digital Futures Seminar, KTH Royal Institute of Technology, Stockholm, Sweden, Mar. 2026.
4. "Beyond AdamW: Developments in Training Algorithms for Deep Learning". MS&E Seminar, Stanford University, Palo Alto, California, Feb. 2026.
5. "Purifying Shampoo: Understanding and Improving Shampoo Through An Approximation Perspective". INFORMS Annual Meeting, Atlanta, Georgia, Oct. 2025.
6. "Scaling Beyond Diagonal Preconditioners for Training Neural Networks At-Scale". SIAM Annual Meeting, Montreal, Canada, July 2025.
7. "Going Beyond Diagonal Preconditioners for Training Neural Networks At-Scale". Flatiron Institute, Center for Computational Mathematics, Remote, Nov. 2024.
8. "Going Beyond Diagonal Preconditioners for Training Neural Networks At-Scale". University of Zagreb, Zagreb, Croatia, Oct. 2024.
9. "Going Beyond Diagonally Preconditioned Methods for Training Neural Networks At-Scale". LinkedIn, Sunnyvale, California, May 2024.
10. "Going Beyond Diagonal Preconditioners for Training Neural Networks At-Scale". LANS Seminar, Argonne National Laboratory, Lemont, Illinois, Apr. 2024.
11. "Going Beyond Diagonal Preconditioners for Training Neural Networks At-Scale". Amazon AI AWS Labs, Dec. 2023.
12. "Distributed Data-Parallel Training of Neural Networks At-Scale Using Distributed Shampoo". SIAM Conference on Optimization 2023, Seattle, Washington, June 2023.
13. "Distributed Data-Parallel Training of Neural Networks At-Scale Using Distributed Shampoo". FAIR Montreal, Montreal, Canada, April 2023.
14. "Adaptive Finite-Difference Methods for Noisy Derivative-Free Optimization". INFORMS 2021, Remote, Oct. 2021.
15. "Stochastic Line Searches (@FB?)". Facebook, Menlo Park, California, Sept. 2021.
16. "Finite-Difference Methods for Noisy Derivative-Free Optimization". Math 290J, UCLA, April 2021.
17. "Recent Advancements in Stochastic, Noisy, and Derivative-Free Optimization Methods for Machine Learning". William and Mary, Feb. 2021.
18. "A Noise-Tolerant Quasi-Newton Method for Unconstrained Optimization". INFORMS 2020, Remote, Nov. 2020.
19. "Compositional Embeddings Using Complementary Partitions for Memory-Efficient Recommendation Systems". ACM SIGKDD, Remote, Aug. 2020.
20. "Compositional Embeddings Using Complementary Partitions for Memory-Efficient Recommendation Systems". PeRSoNAI tutorial, ISCA 2020, Remote, May 2020.
21. "Towards Understanding Embeddings and Optimization in Deep Neural Recommendation Systems". INFORMS 2019, Seattle, Washington, Oct. 2019.
22. "Recent Advancements in Stochastic Quasi-Newton Methods". Facebook, Menlo Park, California, Feb. 2019.
23. "A Progressive Batching L-BFGS Method for Machine Learning". Chicago Area SIAM Student Conference 2018, Chicago, Illinois, Apr. 2018.

24. "A Progressive Batching L-BFGS Method for Machine Learning". Midwest Machine Learning Symposium 2018, Chicago, Illinois, June 2018.
25. "Learning Near-Isometric Linear Embeddings". Joint Mathematics Meetings 2015, San Antonio, Texas, Jan. 2015.

## Teaching Experience

**Northwestern University**

- IEMS 490: Seminar Topics in Optimization for Machine Learning, Guest Lecturer (Spring 2024)
- IEMS 351: Optimization Methods in Data Science, Instructor (Spring 2020)
- IEMS 455: Machine Learning, Teaching Assistant (Spring 2018)
- IEMS 1st Year Boot Camp: Analysis, Instructor (Summer 2017)

**University of Maryland**

- CMSC 828G: Systems for Machine Learning, Guest Lecturer (Spring 2025)

## Advising

**Graduate Students and Research Interns Advised**

- Kang An, Rice University, Summer 2026 (co-advised with Anna Cai).
- Hiroki Naganuma, Université de Montréal and Mila, Summer 2025 (co-advised with Parameswaran Raman).
  Next Position: Deep Learning Software Engineer at NVIDIA.
- Runa Eschenhagen, University of Cambridge, Summer 2024 – Winter 2025.
  Next Position: Member of Technical Staff at Anthropic.
- Jose Gallego-Posada, Université de Montréal and Mila, Winter 2023 – Winter 2024 (co-advised with Michael Rabbat).
  Next Position: Quantitative Researcher at The Voleon Group
- Anna Cai, Carnegie Mellon University, Summer 2023.
  Next Position: Software Engineer at Meta Platforms.
- Manish Kumar, Northwestern University, Winter 2020.
  Next Position: Data Scientist at Microsoft.

**Committee Member / Thesis Opponent**

- Zesen Wang, KTH Royal Institute of Technology, March 2026.

## Professional Activities

- Referee for:
  - *Journal of Machine Learning Research*
  - *SIAM Journal on Optimization*
  - *Mathematical Programming*
  - *Mathematical Programming Computation*
  - *Computational Optimization and Applications*
  - *Journal of Optimization Theory and Applications*
  - *International Conference on Machine Learning*
  - *Neural Information Processing Systems*

  - *International Conference on Learning Representations*
  - *The International Conference for High Performance Computing, Networking, Storage, and Analysis*

- Co-Organizer, AlgoPerf Workshop (MLCommons), Menlo Park, CA, Feb. 2025.
- SIAM Member.
- INFORMS Member.
- Northwestern INFORMS Student Chapter Board: Webmaster (2017-18).

## References

**Jorge Nocedal** (Ph.D. Advisor)
Professor, Industrial Engineering and Management Sciences
Northwestern University
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**Michael Rabbat**
Co-Founder and Vice President of World Models
AMI Labs
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**Maxim Naumov**
Director of Engineering
Core ML, Google
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